Selected normalization of character traces

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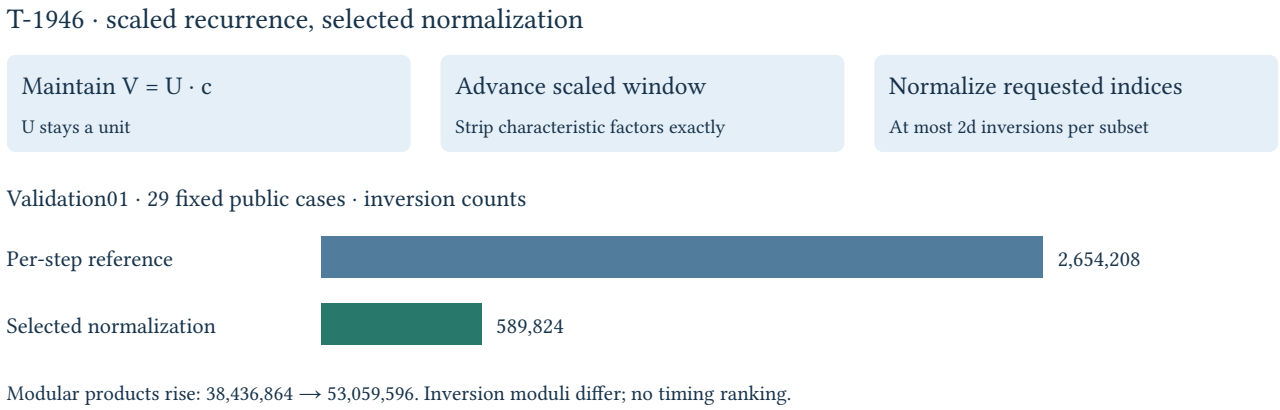
Construction. Retain V=U·c for the coefficient window, with U a unit. At step k, strip the characteristic factors from the recurrence numerator and working modulus. Scale surviving entries by the remaining denominator factor and update U. Invert U modulo r only when the coefficient contributes to a requested trace. Constant coefficient requires no inverse.



Exact checks. 7,965 coefficients in45 polynomial controls,90 traces and all36 word systems match. Nine cap decisions and their costs are retained. Fourteen singleton phases checked. Public29 outcomes match:12 extracted,11 identity/endpoints and6 subset caps. Steps, calls and precision boundaries agree. Empty request, malformed index and both large preflight refusals pass. No native compile/assertion failure.

Accounting. Trace inversions648 versus15,840; products284,086 versus210,078. All word systems including refusals:184,722 inversions versus2,764,200;48,651,090 products versus35,982,184. Public selected products53,059,596 versus38,436,864. Selected inversions are in F_r; reference inversions use the current lifted ring. These counts support no timing or total-work ranking. Each public arm retains2,654,208 steps,332 nominal+27 guard queries and19,844 correction multiplications; combined718 queries excluding setup.

Bound and state. Two trace index sets have at most2d distinct positive positions, or d for L1, so normalization uses at most2d inversions per degree-d subset. Request-set/window storage is O(d) entries; precision, containers and source/phase circuits are additional. All coefficient steps and explicit subsets remain. Caps500,000 steps/4,096 subsets/native modulus ceiling unchanged.

Custody and next. Thirteen copied files/17 dependencies hash-bound. Original4,563-case cohort and prior code unchanged. Next nonsingular matrix blocks with explicit singular boundaries and requested checkpoints; no block jumps are implemented here. The whole-word subset representation remains a separate dependency. Source, set lookups, reductions, Euclidean work, allocation and verification remain charged.

Sources. T1944 precision proof/source and Harvey–Sutherland II section2 equations2–4/remark2.2. Common-denominator formulation is established; no novelty or many-prime bound imported. KB unavailable. Independent signed review pending; no cloud work.